Anatomical Footprint of the Impulse Control Disorders in Parkinson's Disease: a Convolutional Vision Transformers Approach

Edwin Valarezo Añazco[1,2], Carlos Jiménez F.[1,3], Alisson Constantine M.[1,2], Enrique Peláez[1,2], Francis R. Loayza[1,3*], Ignacio Obeso[4], and Gianpaolo Maggi[5]

[1] Escuela Superior Politécnica del Litoral, ESPOL, Campus Gustavo Galindo Km 30.5 Vía Perimetral, P.O. Box 09-01-5863, 090150 Guayaquil, Ecuador.
[2] Facultad de Ingeniería en Electricidad y Computación (FIEC)
[3] Facultad de Ingenieria en Mecánica y Ciencias de la Producción (FIMCP)
[4] HM CINAC (Centro Integral de Neurociencias Abarca Campal), Hospital Universitario HM Puerta del Sur, HM Hospitales, Madrid, Spain
[5] Department of Psychology, University of Campania "Luigi Vanvitelli", 81100 Caserta, Italy
{edgivala,cajifarf, alascons, epelaez, floayza}@espol.edu.ec, {iobeso}@ucm.es, {gianpaolo.maggi}@unicampania.it
http://www.espol.edu.ec/es

Abstract. Impulse Control Disorder in Parkinson's patients are a psychiatric condition characterized by pathological gambling, compulsive buying, binge eating and hypersexuality. Even though that Impulse Control Disorders are attributed to dopaminergic treatment, recent studies associate this condition with abnormalities in brain anatomy. In this work, we used T1-weighted magnetic resonance images of three groups of Parkinson's Disease subjects: 1) idiopathic Parkinson's disease, 2) Pakinson's patients with hypersexuality and eating impulsive control disorders, and 3) Parkinson's patients presented compulsive buying and gambling. All the images were pre-processed including bias field correction, denoising, registration to the MNI template, and brain extraction. To identify each condition, transformer-based deep learning models were used. The Convolutional vision Transformer model uses 3D-convolutional layers to recognize patterns of Impulse Control Disorders in brain images. The 3D-convolutional layers were utilized to analyze the complete 3D brain anatomy and the larger receptive field as compared to 2D convolutions. Additionally, a ResNet model was implemented to compare performances. The ResNet model was based on connections between non-consecutive 3D convolutional layers. Testing results show that the transformer model was more efficient in terms of memory usage, training time, and classification accuracy compared to the ResNet model.

Keywords: Parkinson · Deep Learning · Vision Transformers · Impulse Control Disorder

* Corresponding Author

1 Introduction

Impulse Control Disorders (ICDs) are a prevalent group of psychiatric disorders characterized by the inability to resist urges or impulses, leading to maladaptive and self-destructive behaviors [28].

In patients with Parkinson's Disease (PD), the incidence of Impulse Control Disorders (ICDs) has been observed to be higher as compared to the general population [8, 22]. This increased prevalence has notably compromised the overall quality of life and well-being of the affected patients [24]. Although the neurobiological basis of ICDs in PD remains elusive, recent evidence suggests that alterations in the structure of brain anatomical patterns within specific brain regions might be related to the development of these disorders [9, 15, 19, 29].

Deep Learning (DL) methodologies, in particular those utilizing Transformer architectures, have revolutionized the field of Artificial Intelligence (AI) and have demonstrated remarkable performance in various domains, including vision [1], natural language processing, and more recently, biomedical research [2, 12, 12], brain tissue segmentation [20, 21] and others.

In this paper, we propose a novel approach to recognize ICD patterns in Parkinson's patients using a transformer-based model. The proposed model's architecture was designed using 3D-convolutional layers to be able to analyze the complete 3D brain and identify patterns that indicate ICDs in PD patients. The Convolutional vision Transformer (CvT) model was evaluated using the Parkinson Progression Marker Initiative (PPMI) public database; its performance was compared against a classifier based on residual connections (i.e., ResNet model) in terms of computational efficiency and classification accuracy. The results showed that the CvT model outperforms the ResNet model in all experiments.

2 Related work

The transformer architecture has shown encouraging results in natural language processing and computer vision tasks [13]. Recently, researchers have explored this model in automated recognition of PD-related brain regions. Zhang et al. [31] in the supplement material examined a wide variety of DL techniques used in radiomics medical studies in different modalities, types of images, and the models' architectures, with promising results in the diagnosis, visualization, and interpretability of the recognized patterns.

DL techniques have also been used to recognize and locate regional patterns in different types of images. Wang et al. [30] proposed using the transformers' multi-level attention mechanism to automatically identify a disease-related region in a Computed Tomography (CT) image and annotate the patterns, combining clinical knowledge and images, resulting in significant disease classification.

While the use of radiomic analysis in combination with DL methods has shown promising results in disease diagnosis [31], recent research is also exploring the use of DL Transformers to categorize and locate lesions using different

types of data, such as in [7,14]. Chetoui and Li et al. [7,14] have shown promising results in proper prognosis and prevention of local treatment failures by recognizing patterns associated with high-risk patients with early manifestations of symptoms, such as the ICDs in parkinsonism.

Zhang et al. [31] in their study also analyzed the use of DL models in conjunction with extracted feature analysis to learn phenotypic features from 2D and 3D images, without predefined characteristics or human intervention. The authors discussed extracting radiomic features from CT scans and used Convolutional Neural Networks (CNNs) to identify and classify brain tumors based on their radiomic features. The authors noted that this approach required less training time and was able to identify more relevant regional features for disease diagnosis than previous models.

Yan Han et al. [10] proposed a Radiomics-Guided Transformer (RGT) which combines the transformers architecture for image analysis and radiomics information to identify, localize and classify cardiopulmonary pathologies without any bounding box annotations. The authors used 112.120 chest X-rays images labeled with one or more of 14 cardiopulmonary diseases collected from 30.805 subjects. The proposed approach outperformed weakly supervised learning methods for disease localization.

Qing Lyu et al. [16] also developed a transformer model to classify tumors from a whole-brain MRI, using 582 contrast T1-weighted images used for segmentation, location, and then its classification into one out of five classes. The proposed architecture outperformed known models with an Area Under the Curve (AUC) of 0.878. The authors showed that with the whole-brain imaging features, the model could diagnose the presence of tumor patterns, locate the organ site and classify a metastatic tumor type using an MRI of the whole brain.

Although, several studies analyze the non-motor symptoms of PD using MRI, there is no progression tools to allow the tracking of ICDs within PD over time. Today, the assessment of PD progression in conjunction with ICD is based on patient self-report or a physician's qualitative rating.

3 Materials and Methods

3.1 Database

MRI images were obtained from the PPMI database. A total of 600 images were extracted from patients using the following criteria:

I. Presence of at least two of the three cardinal signs of Parkinson's disease (i.e., bradykinesia, rigidity, or resting tremor).
II. Availability of neuropsychological assessment data. Patients were classified using the criteria described in [17].
III. Absence of clinical diagnosis of dementia in the patients.

As a result, the following groups were extracted: 35 subjects reported pathological gambling ($PD + ICD_G$); 63 subjects with compulsive sexual behavior

($PD + ICD_X$); 126 subjects presented compulsive shopping ($PD + ICD_S$); 261 subjects with an eating disorder ($PD + ICD_E$); and 115 PD patients without ICDs (only PD). In addition, all participants provided a written informed consent, and each site of the PPMI database received approval from an ethical standards committee on human experimentation. Data from the PPMI database was accessed on the 12th of February 2021.

3.2 Pre-processing Protocol

Anatomical T1-weighted images were pre-processed using a Python script based on the ANTS library [5], running on a 12-core CPU server with 64 Gb RAM and Nvidia GPU A100. All images were pre-processed with a bias field correction using the N4 retrospective algorithm, based on non-parametric non-uniform normalization with B-spline modeling of the bias field [27]. Denoising was also performed using a Gaussian Spatially Adaptive Filter [18]. Images were also registered using the MNI registration process, utilizing a rigid body to avoid non-linear deformations and maintained the native brain shape [6]. All images were brain extracted using the ANTS library, and images were normalized using intensity normalization based on least squares tissue normalization [25].

Data augmentation techniques were performed to increase the number of samples and balance the classes. These augmentation techniques included image rotation, translation, and scaling.

3.3 Model development

ResNet-based Classifier ResNets are a type of CNNs that have shown promising results in image classification tasks [11]. The main idea behind ResNets is using residual connections, which allow the network to learn residual mappings instead of attempting to learn the previous feature map directly. ResNet introduces skip connections that bypass one or more convolution layers, enriching the training by propagating the features and the training error in the forward and backward pass, respectively [4].

Figure 1 shows the architecture of the ResNet model used to classify the 3D MRI images. It uses 3D convolutions because of the larger receptive field as compared to 2D convolutional layers [23]. The ResNet model has been implemented using a 3D convolution layer at the input of the network, with 32 kernels. Then 16 layers were stacked in blocks called ResBlocks, followed by an AveragePool3D layer, two dense layers with 256 neurons each. At the output of the model, an output layer with a SoftMax discriminant function was used to estimate the probability for each class.

There are two types of ResBlock blocks, which are detailed in Table 1:

Convolutional Vision Transformer (CvT) based classifier CvT is a vision model that combines the attention-based modeling of transformers with the

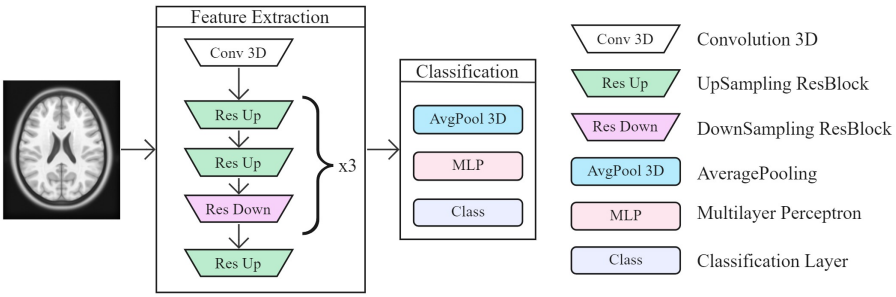


Fig. 1: ResNet model architecture

Table 1: Types of Resblock

Block Type	**Downsample Block**	**Upsample Block**
Description	- Conv3D - BatchNorm3D - Conv3D - BatchNorm3D - ReLU - Conv3D - BatchNorm3D - ReLU - Shortcut - ReLU	- Conv3D - BatchNorm3D - ReLU - Conv3D - BatchNorm3D - ReLU - Shortcut - ReLU

spatial shift-invariance of CNNs [26]. The CvT architecture has shown promising results in various computer vision tasks, such as image classification, object detection, and segmentation [11]. Generally, the input image is fed through convolutional layers to extract local features, then passed to the transformer-based layers which use multi-head self-attention (MSA) to capture global contextual information. The final output is obtained by passing the concatenated feature maps from the transformer and convolutional layers through a set of fully connected layers. The self-attention mechanism is computed as follows:

$$Attention(Q, K, V) = Softmax(\frac{QK^T}{\sqrt{d_k}})V \tag{1}$$

where Q, K and V are the Query, Key and Value matrices respectively and d_k is the dimension of the input vectors. All of which are vectors derived from the input of the transformer block [3].

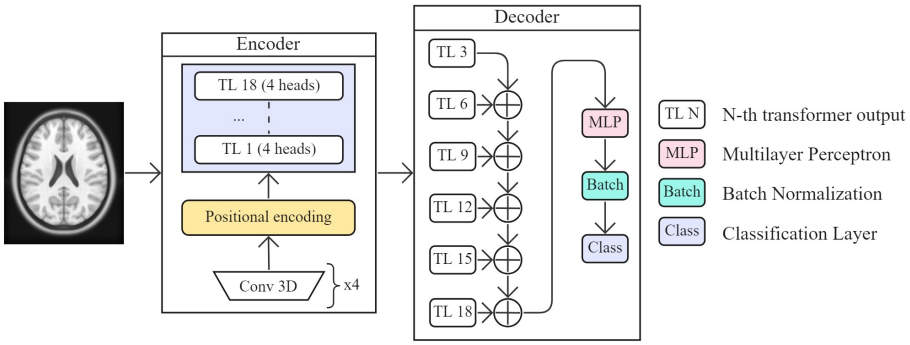


Fig. 2: CvT model architecture

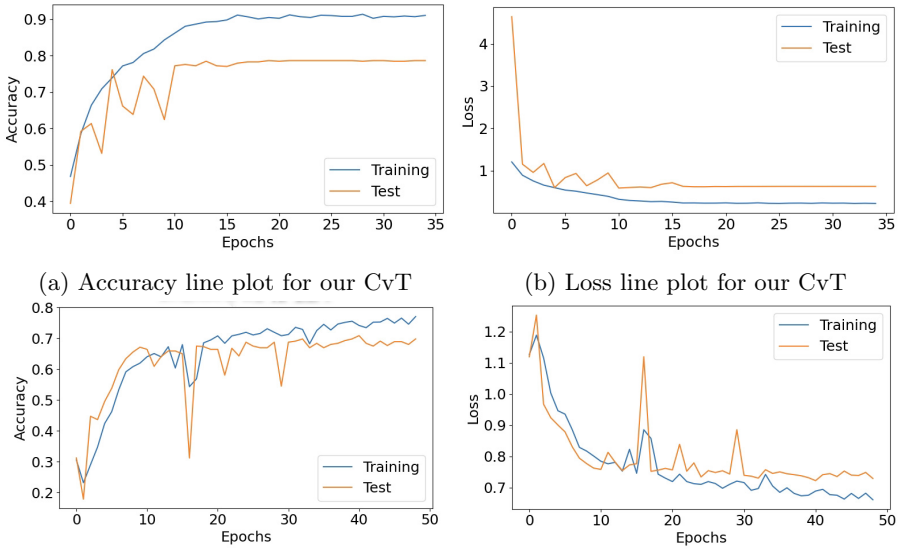
The architecture of our proposed CvT model is shown in Figure 2. It comprises an encoder - decoder design pattern. The encoder has four 3D convolutional layers with a kernel of 3x3x3. Then, 18 transformer blocks with MSA layers with 4 heads each, 2 fully connected layers of 256 neurons each, and 0.15 of dropout. All convolutional layers of the encoder uses the RELU activation function. As for the decoder, it concatenates the intermediate outputs (layers: 3, 6, 9, 12, 15, 18) of the transformers. Batch normalization was also performed, followed by a Multi-layer Perceptron (MLP) layer with 128 neurons and a ReLu activation function. The output layer has a fully connected layer with three neurons and a SoftMax activation function.

3.4 Experiments

The experiments for both DL-based models were run on an NVIDIA RTX 3090, with 24GB VRAM GPU using Tensorflow 2.11.0, Pytorch 1.13.1+cu117, and Python 3.9.

Training and validation The ResNet model was trained using a cross-entropy loss function for multi-class classification, and Adam optimizer.The CvT-based model was trained using a cross-entropy loss, Adam optimizer, and a batch size of 16.

We performed a 5-fold cross-validation test to evaluate our models, which is helpful when working with relatively small datasets. Training and testing datasets were randomly created, splitting the data into five equal parts, leaving 1/5 of it (i.e., 20%) out for testing and the rest for training. This process was repeated five times, so the model was evaluated with five different testing sets during each iteration.



(a) Accuracy line plot for our CvT

(b) Loss line plot for our CvT

(c) Accuracy line plot for our ResNet

(d) Loss line plot for our ResNet

Fig. 3: Loss and accuracy line plots for both DL models

Performance evaluation The performance of the models were also assessed using the area under the receiver operating characteristic curve (AUC-ROC) and accuracy metrics. The equations of these performance metrics are shown below:

$$AUC - ROC = \frac{TP}{TP + FN} \times \frac{TN}{TN + FP} \tag{2}$$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{3}$$

where TP is the true positive, TN is the True Negatives, FP is the False Positives, and FN is the False Negative.

4 Results

This section presents the classification results obtained by the proposed CvT and ResNet-3D models. For the experiments, three classes were created.

a) PD patients with hypersexuality and eating disorders were grouped (i.e., $PD + ICD_X + ICD_E$);

b) PD patients with gambling with shopping disorders (i.e., $PD + ICD_G + ICD_S$); and,

c) PD patients without any disorders (i.e., PD).

The classes were created based on the hypothesis that the disorders share similar brain anatomy structures and activate similar regions. Table 2 shows

the training accuracy and loss, the validation accuracy and loss, and the ROC-AUC scores computed from the evaluation process. Our proposed CvT model achieved an accuracy value of 77% and 0.9828 of ROC-AUC. The ResNet-3D model reached 70% accuracy and a ROC-AUC score of 0.8459.

Table 2: Performance evaluation of our models.

Model	***Acc (%)***	***loss***	***Val. Acc (%)***	***Val. Loss***	***ROC-AUC***
CvT	**86**	0.32	**77**	0.58	**0.9828**
ResNet3D	74	0.68	70	0.72	0.8459

Figure 3 depicts the training process as recorded in each training cycle, showing the accuracy and loss values for our proposed DL models. As observed in Figures 3a and 3b, during the initial training phase, both the loss and accuracy oscillate until the CvT model stabilizes at approximately 18 epochs, reaching its highest accuracy value of 77% with the validation dataset and 86% with the training dataset.

3c and 3d show both the accuracy and loss of the ResNet proposed model. The accuracy fluctuates throughout the training process and gently stabilizes after epoch 40. In addition, both curves display erratic behavior, presenting several spikes that deviate from the underlying trend of the curve, particularly around epochs 17 and 29.

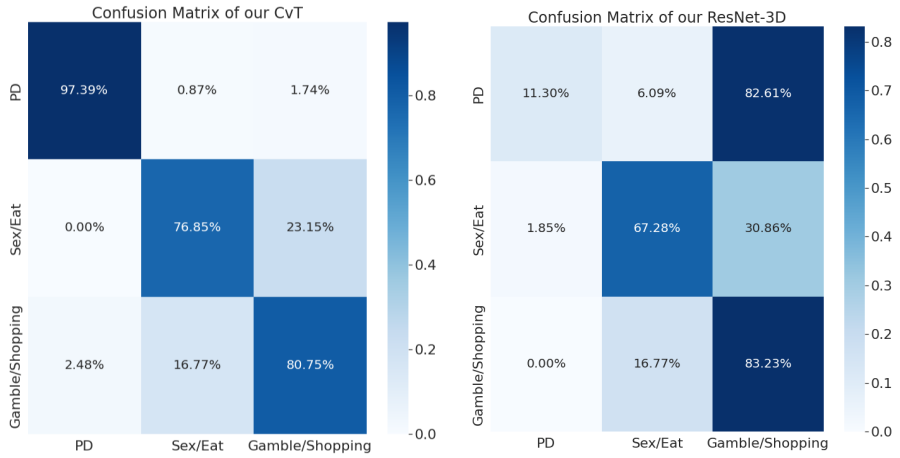
The average confusion matrices with the training dataset for CvT and ResNet models are presented in Figure 4, rows correspond to ground-truth labels and columns to predicted labels. Figure 4a shows that the CvT model has a precision of 97% for Parkinson's patients (i.e., PD), 77% for *Sex and Eat* class (i.e., $PD + ICD_X + ICD_E$), and 81% for the *Gamble and Buy* (i.e., $PD + ICD_G + ICD_S$). The larger confusion is presented between classes $PD + ICD_X + ICD_E$ and $PD + ICD_G + ICD_S$).

Figure 4b shows the results of the ResNet-3D model. This architecture was not able to correctly distinguish between the three classes. For instance, PD has a precision of 11.3%, and 82.61% of the PD class were classified $PD + ICD_G + ICD_S$.

It is also worth mentioning that both DL models consumed all the GPU memory available (i.e., 24GB). However, our CvT was able to complete the training process with a larger batch size (16), meanwhile, our RestNet-3D model only allowed two training samples per batch. Regarding training time, our CvT completed, on average, seven epochs per hour, whereas ResNet completed 3.47 epochs per hour, making the CvT training faster.

5 Discussion

In this work, we analyzed the anatomical difference between PD patients and PD patients with ICDs (i.e., pathological gambling, hypersexuality, shopping, and



(a) Confusion matrix for CvT (b) Confusion matrix for ResNet-3D

Fig. 4: Confusion matrices of both DL models

eating disorders) from T1-weighted MRI images. The classification performance of the Convolutional vision Transformer was compared against a ResNet-based model, which represents the state of the art in some computer vision tasks.

The experiments showed that our proposed CvT model better recognized Parkinson's patients and Parkinson's + Disorders patients than the ResNet model, achieving an average accuracy of 77% and 70%, respectively. Additionally, the confusion matrix of the CvT model showed fewer misclassification errors than the confusion matrix of the ResNet model. Thus, the CvT model better learns the anatomical patterns of the patient's brain to distinguish between the classes. The ResNet model achieved an AUC score of 0.8459, showing a reduction in the True Positives and an increment in the False Positives of each class as compared to the CvT model.

Regarding the training time efficiency, the CvT had a faster convergence time than the ResNet model, attaining its highest accuracy after about 5 hours, in contrast to the ResNet-3D, which showcased its peak accuracy in approximately 15 hours.

These results suggest that the CvT architecture of the transformer efficiently learned the 3D brain-MRI patterns to distinguish correctly between disorders in less training time and fewer training iterations.

Running the experiments, it was noticed that the CvT model was 8x more memory-efficient than the ResNet-3D. It is important to point out that the models can be further improved with more computational resources. However, ResNet will demand more training time and memory usage than the CvT model.

An important drawback of Tranformer-based Networks is their voracious data requirements. Typically, the accuracy and precision of Transformers predictions most likely improve when using larger data sets. Acquiring more data for each

class is a crucial task that should be addressed in order to expect more promising results. Furthermore, acquiring more data will add more complexity to the network, such as memory consumption and processing resources for training. The models implemented in this research will require more powerful GPUs to overcome hardware limitations during the experiments.

6 Conclusions

In this research, two Deep Learning-based models were implemented and evaluated to recognize ICD in Parkinson's patients using 3D MRI images. First, the Convolutional vision Transformer model was built with 3D-convolutional layers because the larger receptive field of 3D convolutions allows the model to extract features from the complete 3D brain anatomy. Second, the ResNet model uses skip connections to propagate the features in the forward pass and backpropagate the training error, avoiding vanishing it.

Results from experimentation revealed that the proposed CvT model outperformed the ResNet model in recognizing ICD patterns of PD patients using MRI images. The CvT's accuracy was 77% as compared to the 70% accuracy reached by the ResNet model. There was also a better ROC-AUC score (0.9828 versus 0.8459). The CvT model also showed fewer miss-classification errors than the ResNet model. For instance, 23.15% of the class $PD + ICD_X + ICD_E$ were misclassified as $PD + ICD_G + ICD_S$ with CvT model, meanwhile 82.61% of the PD class were classified $PD + ICD_G + ICD_S$ with the ResNet model. Thus, The results suggest that CvT model has learned the anatomical patterns of the patient's brain more effectively to distinguish between the classes.

The CvT model converged faster and had a more stable training curve than the ResNet-3D model. Actually, the CvT model needs less training time to reach peak accuracy compared to the ResNet-3D. This implies that the CvT architecture is trained in less time and with fewer iterations. Then, the CvT training could be scaled to larger datasets.

These findings could contribute to better diagnostic processes and intervention strategies for patients with PD suffering from ICDs. Although both models consumed the available GPU memory, the CvT model demonstrated better memory utilization efficiency, potentially leading to less resource demand in future iterations or model improvements.

This study grouped ICDs based on the hypothesis that similar disorders share similar brain pathways related to similar regions. This could help further understand these disorders' neurological basis in PD patients. Future studies will elaborate on using explainability algorithms and statistical analysis to show the anatomical differences used by the CvT model to classify the patients.

7 Acknowledgment

Data used in the preparation of this article were obtained from the Parkinson's Progression Markers Initiative (PPMI) database (www.ppmi-info.org/access-data-

specimens/download-data). For up-to-date information on the study, visit www.ppmi-info.org.

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